

Agricultural Agent

A specialized system that provides crop recommendations by analyzing climate conditions, soil types, and suitable crops.

Agent 1: Climate Analysis Agent

Evaluates historical weather patterns for a region including temperature ranges, precipitation levels, humidity, and growing season length. Requires location information (region, coordinates, or area name) from the user to create a climate profile that is forwarded to the next agent.

Agent 2: Soil Recommendation Agent

Determines optimal soil types for the given climate conditions, considering drainage requirements, soil composition, and pH levels. Uses the climate profile to generate soil recommendations that are passed to the Crop Selection Agent.

Agent 3: Crop Selection Agent

Identifies crops that would thrive under the specified climate and soil conditions, categorizing them by type and considering water requirements, temperature tolerance, and growth cycles.

Agent 4: Pest and Disease Management Agent

Identifies potential pest and disease threats based on climate conditions and historical patterns, suggesting preventive measures and treatment options with a focus on organic approaches. Provides pest management strategies based on the selected crops and regional conditions.

Agent 5: Farming Practice Agent

Develops specific farming guidelines including planting times, spacing requirements, irrigation needs, and fertilization strategies based on the selected crops and their growing conditions.